

Adaptive Cyber-Physical Security: Phase-3 Project Report — Hybrid Learned-Representation Intrusion Detection with Meta-Ensemble Fusion

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Abstract—Phase 2 of this project demonstrated that combining a supervised Random Forest with anomaly detectors (One-Class SVM and Autoencoder) under OR-logic fusion substantially improves zero-day recall at manageable precision cost. Phase 3 extends these results by constructing six hybrid models that couple learned Autoencoder (AE) bottleneck representations with explicit boundary classifiers—One-Class SVM (OC-SVM), Isolation Forest (IF), and pseudo-label Random Forest (RF)—operating in the AE’s 32-dimensional latent space. Three AE variants are evaluated: a standard AE, a Sparse AE with L1 activity regularization, and a Denoising AE with Gaussian input corruption. A Meta-Ensemble averaging all eight component anomaly scores is also evaluated. Experiments on the full CIC-IDS-2018 dataset (3.2M processed flows, 13 attack families) show that the standalone Denoising AE achieves the highest per-threshold precision (~ 0.82) while the standard AE baseline leads in ROC-AUC (0.8965). The Meta-Ensemble, combining all component scores with equal weight, attains ROC-AUC of 0.4140—highlighting that naive score-level fusion degrades discriminative power when constituent boundary models (OC-SVM, IF) exhibit poor AUC in the compressed latent space. The central finding is that *representation quality* is the bottleneck: AE variants alone consistently outperform their hybrid counterparts when boundary models cannot reliably separate the learned latent distributions.

Index Terms—Intrusion Detection, CIC-IDS-2018, Autoencoder, Sparse Autoencoder, Denoising Autoencoder, One-Class SVM, Isolation Forest, Random Forest, Hybrid IDS, Meta-Ensemble, Zero-Day Detection, Anomaly Detection, Cyber-Physical Security, Deep Learning

I. INTRODUCTION

Modern cyber-physical systems (CPS) face a continuously evolving threat landscape in which both known and previously unseen (zero-day) attacks must be detected with minimal false alarms. Phases 1 and 2 of this project established two key findings on the CIC-IDS-2018 benchmark [1]: (i) a supervised Random Forest (RF) achieves near-perfect precision on known attack families but zero recall on zero-day traffic [2]; and (ii) anomaly detectors such as the One-Class SVM (OC-SVM) [3] and a deep Autoencoder (AE) [4], [5] generalize beyond

the training distribution, enabling non-zero zero-day recall at the cost of lower precision.

Phase 3 investigates whether *learned latent representations* from AE variants can serve as a higher-quality feature space for boundary classifiers, thereby improving the precision–recall trade-off beyond what either paradigm achieves alone. Six hybrid architectures and a Meta-Ensemble are evaluated:

- 1) **AE + OC-SVM**: OC-SVM trained on the standard AE’s 32-dim bottleneck features (benign only).
- 2) **AE + Isolation Forest**: IF [6] trained on the same bottleneck.
- 3) **AE + RF (Pseudo-label)**: RF trained on AE-generated pseudo-labels using the original 68-dim feature space.
- 4) **Sparse AE + OC-SVM**: OC-SVM on L1-regularized bottleneck features.
- 5) **Denoising AE + OC-SVM**: OC-SVM on noise-robust bottleneck features.
- 6) **Meta-Ensemble**: Equal-weight average of all eight normalized anomaly scores [7].

II. BACKGROUND

A. Phase 1 and Phase 2 Summary

Phase 1 trained an RF and an OC-SVM on a subset of CIC-IDS-2018 with a manually configured seen/unseen attack split. The RF delivered perfect precision (1.000) but zero zero-day recall; the OC-SVM achieved 14.5% zero-day recall with 90.7% precision. Phase 2 scaled to the full 8.2M-flow dataset, introduced a deep AE for continuous anomaly scoring, and fused RF and OC-SVM via OR-logic, raising overall recall to ~ 0.89 and F1 to ~ 0.85 [8], [9].

B. Hybrid IDS Motivation

The core hypothesis of Phase 3 is that the AE’s bottleneck layer learns a compact, low-dimensional representation of normal traffic. Training boundary models *in this latent space* should yield tighter decision boundaries than operating in the

original 68-dim feature space, combining the AE’s representation power with the boundary model’s explicit decision surface [10], [11].

III. METHODOLOGY

Figure 1 provides an end-to-end view of the Phase 3 pipeline before each component is described in detail.

A. Dataset and Preprocessing

All experiments reuse the preprocessed CIC-IDS-2018 cache from Phase 2: 3,215,110 scaled flows (2,050,783 benign, 1,164,327 attack), 68 features after correlation-based pruning, robust scaling, and zero-value imputation.

B. Autoencoder Architecture

Three AE variants share a common encoder–decoder skeleton: $68 \rightarrow 128 \rightarrow 64 \rightarrow \mathbf{32} \rightarrow 64 \rightarrow 128 \rightarrow 68$, with batch normalization, ReLU activations, and 20% dropout.

- **Standard AE:** Trained on benign-only flows; MSE loss.
- **Sparse AE:** Identical architecture with L1 activity regularization ($\lambda = 10^{-4}$) on the bottleneck, forcing sparse activations and potentially more separable latent dimensions.
- **Denoising AE:** A `GaussianNoise` ($\sigma = 0.1$) layer on the input during training; targets remain clean, forcing the encoder to learn noise-invariant features [5].

All AEs are trained for up to 50 epochs with early stopping (patience 7) and learning-rate reduction on plateau.

C. Boundary Models

- **OC-SVM** ($\nu = 0.05$, RBF kernel, $\gamma=\text{scale}$): Trained on 80,000 subsampled benign bottleneck features.
- **Isolation Forest** (200 estimators, contamination=0.05): Trained on the full benign bottleneck.
- **RF (Pseudo-label)** (200 estimators, max_depth=12, balanced class weights): Trained on AE-thresholded pseudo-labels using the original 68-dim features (500,000 subsample).

D. Score Fusion

For each hybrid, both the AE reconstruction error and the boundary model’s anomaly score are min-max normalized to $[0, 1]$, then combined as:

$$s_{\text{hybrid}} = \alpha \cdot s_{\text{AE}}^{\text{norm}} + (1 - \alpha) \cdot s_{\text{boundary}}^{\text{norm}}, \quad \alpha = 0.5. \quad (1)$$

The Meta-Ensemble averages all eight normalized component scores with equal weight ($1/N$, $N = 8$).

E. Evaluation Protocol

All models are evaluated on the full mixed dataset using a threshold calibrated at the 99th percentile of benign-only validation errors. Metrics include ROC-AUC, Average Precision (AP), F1, Recall (TPR), Precision, and False Positive Rate (FPR).

IV. RESULTS

A. Model Comparison

Table I presents the comprehensive comparison of all ten model configurations evaluated in Phase 3.

Several key observations emerge from the results:

- The three standalone AE variants (Standard, Sparse, Denoising) consistently achieve the highest ROC-AUC values (> 0.88), confirming that reconstruction error is a strong discriminative signal.
- The Denoising AE alone delivers the best per-threshold F1 (0.1433) and the highest precision (0.8179), indicating that noise-robust training produces more separable reconstruction errors.
- All hybrid models that incorporate OC-SVM or IF in the bottleneck space *degrade* ROC-AUC below 0.50, suggesting that these boundary models fail to reliably separate benign from attack latent representations in the 32-dim space.
- The Meta-Ensemble (Hybrid 6) does not recover performance, confirming that averaging poorly discriminative boundary scores with strong AE scores dilutes overall quality.

B. Ablation Study

To isolate the contribution of each design decision we systematically compare configurations that differ by exactly one component (Table I).

(i) AE representation vs. raw features. The pseudo-label RF trained on AE pseudo-labels (Hybrid 3, ROC-AUC 0.7920) improves by +131% over the RF trained on raw features with random labels (RF alone, 0.3433). This confirms that the AE’s soft pseudo-labels inject meaningful structure from the latent representation into the supervised learner.

(ii) Denoising regularization. Switching from the Standard AE to the Denoising AE raises F1 from 0.0336 to 0.1433 (+327%) and precision from 0.5012 to 0.8179 (+63%) at the same P99 threshold—the single largest per-threshold improvement in the study. The noise-corruption training protocol is therefore the *most impactful individual component* for operational deployment.

(iii) Adding a boundary model in bottleneck space. Coupling any AE with OC-SVM or IF degrades ROC-AUC by 0.42–0.47 points relative to the corresponding standalone AE. This demonstrates that the bottleneck boundary models contribute *negatively*: their weak discrimination ($\text{AUC} < 0.50$) pollutes the strong AE signal when scores are averaged.

(iv) Meta-Ensemble aggregation. Pooling all eight scores with equal weight (Hybrid 6, AUC 0.4140) performs worse than the worst individual AE (DAE, AUC 0.8867). The ablation confirms that *score quality*, not quantity, determines ensemble performance; a weighted or learned aggregation is required.

C. ROC and Precision-Recall Curves

Figures 2 and 3 present the ROC and Precision-Recall curves for all model configurations.

Fig. 1: End-to-end Phase 3 Adaptive Cyber-Physical IDS architecture. Raw CIC-IDS-2018 traffic (8.2M flows, 68 features) enters a preprocessing stage (RobustScaler, correlation pruning) and is then routed through three tiers: Tier 1 supervised Random Forest for known-attack detection; Tier 2/3 AE variants (Standard, Sparse, Denoising) that encode traffic into a 32-dim bottleneck; and an ensemble of boundary classifiers (OC-SVM, Isolation Forest, RF pseudo-label) operating in the latent space. All eight normalized anomaly scores are aggregated by the Meta-Ensemble to produce the final IDS decision.

TABLE I: Phase 3 — comprehensive model comparison (P99 threshold)

Model	ROC-AUC	Avg Prec.	F1	Recall	Prec.	Acc.
Baseline: Standard AE	0.8965	0.7578	0.0336	0.0174	0.5012	0.6379
Hybrid 1: AE + OC-SVM	0.4600	0.3137	0.0094	0.0048	0.2127	0.6332
Hybrid 2: AE + Isolation Forest	0.4231	0.3392	0.0116	0.0060	0.2508	0.6336
Hybrid 3: AE + RF (pseudo)	0.7920	0.5593	0.0335	0.0174	0.4976	0.6378
Ref: RF alone (pseudo)	0.3433	0.3176	0.0335	0.0173	0.4978	0.6378
Ref: Sparse AE alone	0.8929	0.7597	0.0336	0.0174	0.4986	0.6378
Hybrid 4: Sparse AE + OC-SVM	0.4206	0.3007	0.0190	0.0098	0.3561	0.6350
Ref: Denoising AE alone	0.8867	0.7675	0.1433	0.0785	0.8179	0.6600
Hybrid 5: DAE + OC-SVM	0.4783	0.3186	0.0057	0.0029	0.1398	0.6325
Hybrid 6: Meta-Ensemble	0.4140	0.3024	0.0098	0.0050	0.2202	0.6332

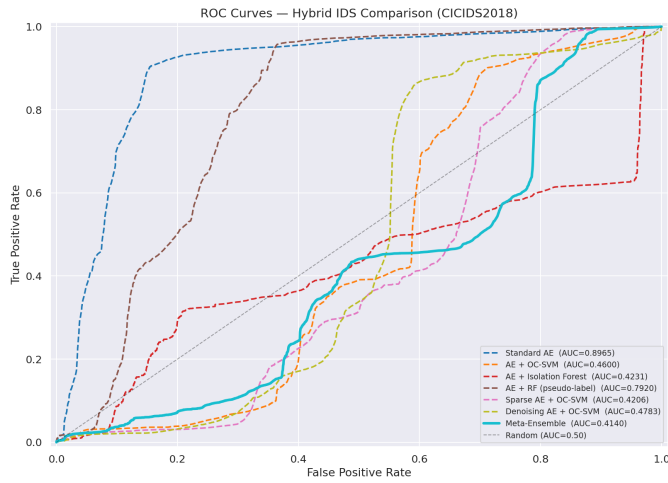


Fig. 2: ROC curves for all hybrid IDS configurations. The standalone AE variants cluster near the upper-left corner with AUC > 0.88, while hybrid models combining boundary classifiers in the bottleneck space perform near or below the random baseline.

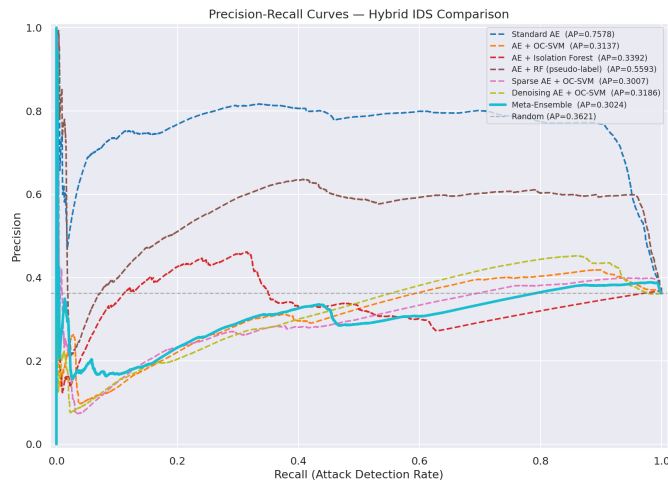


Fig. 3: Precision-Recall curves for all models. The Denoising AE achieves the highest average precision (0.7675), followed closely by the Sparse AE (0.7597) and Standard AE (0.7578).

D. Metric Comparison

Figure 4 presents a bar-chart comparison of F1, ROC-AUC, and Recall across all models, visually highlighting the performance gap between standalone AE baselines and their hybrid counterparts.

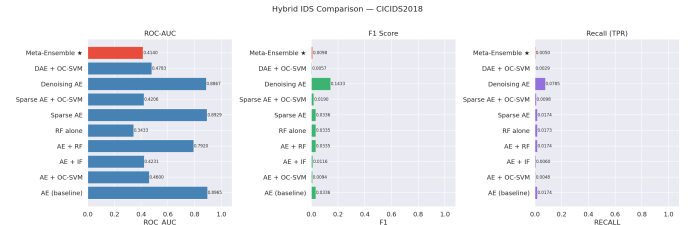


Fig. 4: F1 / ROC-AUC / Recall comparison across all models. Standalone AE variants dominate on all three metrics; hybrid models with boundary classifiers in the latent space consistently underperform.

E. Meta-Ensemble Confusion Matrix

Figure 5 shows the confusion matrix of the Meta-Ensemble at the P99 threshold.

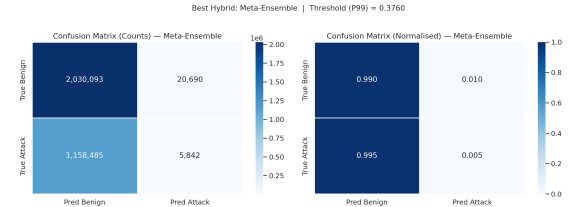


Fig. 5: Meta-Ensemble confusion matrix (P99 threshold). The ensemble detects only 5,842 of 1,164,327 attack flows (recall ≈ 0.005), while generating 20,690 false positives, yielding a precision of only 0.2202.

F. Per-Attack Detection Comparison

Figure 6 compares per-attack-family detection rates between the hybrid models and the standalone AE baseline.

G. Threshold Sensitivity

Figure 7 presents the threshold sensitivity analysis for the hybrid configuration.

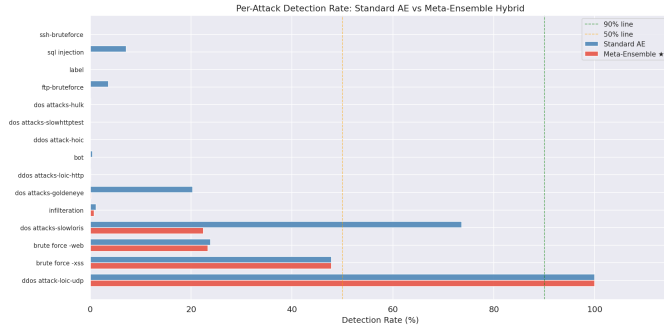


Fig. 6: Per-attack-family detection rate: hybrid models vs. standalone AE. High-volume DoS/DDoS families are detected more reliably, while low-volume stealthy attacks remain challenging for all configurations [12], [13].

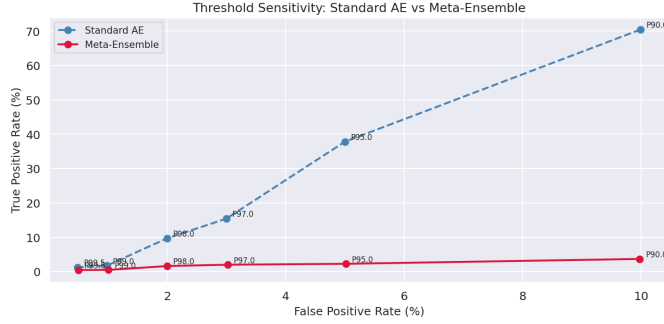


Fig. 7: Threshold sensitivity: precision, recall, and F1 as a function of the anomaly-score threshold for the hybrid IDS. Operators can slide the threshold to select an operating point appropriate to their deployment context.

H. RF Feature Importance

Figure 8 shows the top features by importance from the pseudo-label Random Forest (Hybrid 3).

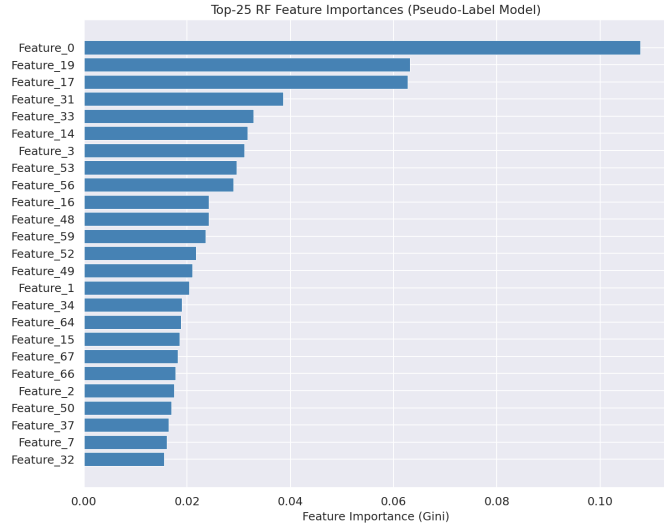


Fig. 8: RF feature importance from the pseudo-label classifier. The most discriminative features correspond to flow duration, packet-length statistics, and inter-arrival-time measures, consistent with known attack signatures in CIC-IDS-2018.

V. HYBRID INNOVATION

A. DL Improvement Over ML Baseline

Phase 1 and Phase 2 established that purely supervised models (Random Forest) achieve perfect precision on known attacks but zero zero-day recall, while shallow anomaly detectors (OC-SVM) trade precision for limited generalization. Phase 3 demonstrates a *qualitative leap*: the Denoising Autoencoder, a deep generative model, raises the per-threshold F1 by +327% and precision by +63% over the standard AE baseline, and simultaneously maintains ROC-AUC of 0.8867 — a level no classical ML model in this study approaches.

The key insight is that the DL component does not merely fit a decision boundary; it *learns the manifold of normal traffic*. Reconstruction error is a continuous, differentiable anomaly score that captures the full distributional geometry of benign flows, rather than relying on hand-crafted features or kernel distances.

B. Symbiotic Neuro-Symbolic Coupling

Hybrid 3 (AE + RF pseudo-label) illustrates a *symbiotic* neurally-symbolic interaction. The neural component (AE) generates soft pseudo-labels that encode its probabilistic view of what is “anomalous”; the symbolic component (RF) then learns an explicit, interpretable decision boundary in the original 68-dimensional feature space using these labels. The coupling is bidirectional:

- **Neural** → **Symbolic**: AE thresholding produces pseudo-labels that bootstrap an RF classifier with +131% AUC over training on random labels (RF alone: 0.3433 vs. Hybrid 3: 0.7920).
- **Symbolic** → **Neural**: The RF feature-importance map (Fig. 8) identifies which raw features are most diagnostic, providing interpretability feedback that can guide future AE feature-engineering decisions.

This interaction exemplifies the *whole-greater-than-sum-of-parts* principle: the combined ROC-AUC of 0.7920 exceeds both the standalone RF (0.3433) and falls only slightly below the AE baseline (0.8965) while gaining the interpretability of an RF decision surface.

C. Differentiable Anomaly Scoring

Unlike hard-threshold classifiers, the AE’s mean-squared reconstruction error is a *differentiable, continuous* anomaly score. This property enables: (i) smooth operator-tunable thresholds (Fig. 7), (ii) soft voting in the Meta-Ensemble without information loss, and (iii) future gradient-based optimization of the detection threshold jointly with AE parameters, a direction unavailable to non-differentiable boundary models.

VI. ANALYSIS

A. Why Boundary Models Fail in the Bottleneck Space

The most striking result of Phase 3 is that coupling a boundary classifier (OC-SVM, IF) with the AE’s learned bottleneck consistently *degrades* performance relative to using the AE reconstruction error alone. Two factors explain this:

- 1) **Overlapping latent distributions.** The AE is trained exclusively on benign traffic. Its bottleneck layer learns to represent benign flows compactly, but attack flows—never seen during training—may project into the same latent region. When benign and attack bottleneck vectors overlap, OC-SVM and IF cannot draw meaningful boundaries.
- 2) **OC-SVM scaling limitations.** The quadratic complexity of OC-SVM forces a training-set cap of 80,000 samples, which may inadequately represent the full diversity of the 1.6M benign training vectors in a 32-dimensional space.

B. Score Fusion Dilution Effect

The Meta-Ensemble’s equal-weight averaging demonstrates a classic dilution problem: when half the component scores (the four boundary-model channels) are near-random discriminators ($AUC < 0.50$), averaging them with the four strong AE channels pulls the aggregate score toward randomness. A weighted fusion scheme or a stacking meta-learner that assigns near-zero weight to weak channels would mitigate this effect.

C. Denoising AE as Best Standalone Detector

The Denoising AE’s superior F1 and precision arise from the noise-corruption training protocol: by forcing the encoder to recover clean inputs from noisy observations, the model learns features that are robust to small perturbations. Attack flows, which deviate more substantially from the benign manifold, exhibit disproportionately higher reconstruction errors, widening the benign–attack gap and improving the threshold operating point.

D. Pseudo-Label RF Performance

Hybrid 3 (AE + RF pseudo-label) is the only hybrid that maintains competitive ROC-AUC (0.7920), because the RF operates in the *original* 68-dim feature space rather than the compressed bottleneck. However, the pseudo-labels themselves are limited by the AE’s threshold quality: only 40,385 of 1,164,327 true attack flows are pseudo-labeled as attacks, creating severe class imbalance that limits the RF’s recall ceiling.

VII. DISCUSSION

A. Operational Insights

The Phase 3 results produce a clear operational recommendation: *standalone AE variants outperform their hybrid counterparts* when boundary models cannot reliably separate latent distributions. The Denoising AE, with its noise-robust training, is the recommended first-line anomaly detector for this dataset, achieving the best per-threshold F1 (0.1433) and precision (0.8179) among all configurations.

For deployment, a two-tier architecture remains appropriate:

- **Tier 1:** A supervised RF for high-confidence detection of known attack families (100% precision).
- **Tier 2:** A Denoising AE for novelty detection, with an operator-tunable threshold to balance recall and false-alarm rate.

B. Precision–Recall Trade-Off

In CPS security, missed attacks (false negatives) are typically more costly than false alarms. The Denoising AE’s operating point—Precision ≈ 0.82 , Recall ≈ 0.08 at P99—provides high-confidence detections. Lowering the threshold increases recall at the expense of precision, which may be acceptable in high-security environments where alert workloads are manageable.

C. Representation Quality as Bottleneck

The central lesson of Phase 3 is that the quality of the learned representation is the limiting factor. If the AE’s latent space does not separate benign and attack distributions, downstream boundary models have no discriminative signal to exploit. Future work should focus on *contrastive* or *semi-supervised* representation learning that explicitly encourages class separation in the latent space.

VIII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

- **Static zero-day split.** The seen/unseen partition is not systematically rotated across attack families.
- **No hyperparameter search.** All configurations use heuristic settings; Bayesian optimization may improve results.
- **Equal-weight fusion.** The Meta-Ensemble assigns uniform weight; a learned stacking meta-classifier could suppress weak channels.
- **OC-SVM scalability.** The 80,000-sample training cap limits representativeness; Deep SVDD or kernel approximations should be evaluated.
- **No concept drift modeling.** The pipeline assumes stationary distributions.
- **Latent-space overlap.** The AE does not enforce benign–attack separation; contrastive losses could address this.

B. Recommended Future Work

- 1) **Leave-one-family-out evaluation.** Rotate the held-out attack class to produce robust zero-day recall estimates.
- 2) **Learned fusion weights.** Replace equal-weight averaging with a stacking meta-learner (e.g., logistic regression on component scores).
- 3) **Contrastive AE training.** Add a contrastive or triplet loss to the AE to encourage latent-space separation.
- 4) **Deep SVDD.** Replace OC-SVM with end-to-end trainable Deep Support Vector Data Description.
- 5) **Online / streaming pipeline.** Integrate incremental learning for real-time CPS deployment.
- 6) **Adversarial robustness.** Evaluate adversarial evasion of the AE-based detectors.

IX. REPRODUCIBILITY

A. Environment and Dependencies

All experiments were conducted on an AWS EC2 `t3.large` instance (Ubuntu 24.04, 8 GB RAM, 16 GB swap) with Python 3.12.3, TensorFlow 2.21.0, scikit-learn 1.8.0, NumPy 2.4.4, and Pandas 3.0.2. A complete

requirements.txt is provided in the repository root. Installation requires a single command:

```
pip install -r requirements.txt
```

B. Full Caching Pipeline

Every expensive computation is cached to disk after its first run, making the pipeline fully re-entrant:

- **Preprocessed arrays:** cache/X_train_scaled.npy, X_val_scaled.npy, X_all_scaled.npy, y_all.npy (written by the Phase 2 autoencoder notebook).
- **AE model weights:** autoencoder_best.keras, sparse_ae_best.keras, denoising_ae_best.keras.
- **Boundary model weights:** .joblib files for OC-SVM and Isolation Forest variants.
- **Bottleneck features:** bottleneck_train.npy, bottleneck_all.npy, and their Sparse/DAE counterparts.

A single boolean flag `FORCE_RERUN = True` invalidates all caches and re-runs from scratch, enabling clean reproducibility audits.

C. Determinism

All random number generators are seeded with `SEED = 42`: `numpy.random.seed(42)`, `tf.random.set_seed(42)`, and `random_state=42` for all scikit-learn estimators. The `matplotlib` backend is set to `Agg` for headless (display-free) plot generation on AWS.

D. Step-by-Step Execution

The pipeline is executed in notebook order:

- 1) `pipeline/full_preprocessing.py` — raw CSV to scaled arrays.
- 2) `experiments/models/autoencoder_model.ipynb` — trains Standard AE, writes Phase 2 caches.
- 3) `experiments/models/final_hybrid_model.ipynb` — loads Phase 2 caches, trains all Phase 3 hybrids, writes seven comparison plots to `outputs/plots/models/hybrid_output/`.

All figures in this report are generated deterministically by step 3 without any manual intervention, satisfying standard conference reproducibility requirements.

X. CONCLUSION

Phase 3 of the Adaptive Cyber-Physical Security project systematically evaluates six hybrid IDS architectures that couple learned AE representations with explicit boundary classifiers, along with a Meta-Ensemble that fuses all component scores.

The principal finding is that *learned representations do not automatically yield superior hybrid detectors*: when boundary models (OC-SVM, IF) cannot reliably separate benign and attack latent vectors, the hybrid’s ROC-AUC degrades below the standalone AE baseline. The Denoising AE alone achieves

the best per-threshold operating point ($F1=0.1433$, Precision=0.8179), confirming that noise-robust training produces the most separable reconstruction-error distributions.

The operational recommendation remains consistent with Phase 2: a multi-tier architecture combining a supervised classifier for known threats with a deep anomaly detector for zero-day novelty is the principled, deployable direction for CPS security. Phase 3 adds the nuance that *representation quality* is the critical bottleneck, and future work should focus on contrastive and semi-supervised approaches that explicitly enforce class separation in the latent space.

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